

Cloth classification using Convolutional neural network

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Abstract—In this project, we explore the application of deep learning techniques for clothing image classification, with a focus on sustainability through the SlowFashion dataset. We begin by experimenting with lightweight models on the Fashion-MNIST dataset, identifying its limitations due to its simplicity. We then move to more complex datasets, including the Clothing Dataset, where we evaluate various convolutional neural network (CNN) architectures. Our study examines the impact of data augmentation, class imbalance handling, and model complexity on classification performance. Notably, we compare standard CNNs with transfer learning approaches using pretrained VGG and ResNet architectures. VGG19, fine-tuned on ImageNet and the Clothing Dataset, achieved the highest accuracy of 70% on the SlowFashion dataset, meeting the target defined by our industry partner. Despite their simplicity, decision trees and small CNNs performed competitively in early experiments, underscoring the importance of appropriate model scaling. We conclude with recommendations for future improvements, including better dataset formatting, the use of top-k accuracy for user interface optimization, and potential benefits from additional fine-tuning on more diverse datasets such as the Fashion Product Images Dataset.

I. INTRODUCTION

Slow Fashion is a start-up that operates a platform for buying and selling second-hand clothing, with a focus on luxury items. Similar in concept to platforms like Vinted, Slow Fashion differentiates itself by offering a secure and user-friendly service, including a premium option in which the company manages the entire selling process on behalf of the user. The platform generates revenue by taking a commission on each sale, a standard model in the resale marketplace. Currently, one major feature offered by competitors but missing from Slow Fashion's platform is automatic categorization of clothing into categories and subcategories. This feature enhances the user experience by streamlining the listing process and improving searchability. The aim of this project is to explore the feasibility of integrating such a feature using artificial intelligence. Specifically, we investigate the use of machine learning models for the automatic classification of clothing into main categories. To this end, we experimented with several publicly available datasets and model architectures, including custom CNNs, decision trees, and state-of-the-art pretrained models like VGG19 and ResNet50. Our goal

was to identify an approach that maximize accuracy, making it suitable for potential deployment in a commercial setting.

II. RELATED WORK

A. *DeepFashion: Powering Robust Clothes Recognition and Retrieval with Rich Annotations* [3]

This paper introduces a large-scale fashion dataset, DeepFashion, which contains over 800,000 images annotated with clothing categories, attributes, and landmarks. The authors propose a unified model based on deep convolutional neural networks (CNNs), specifically employing a VGG-based architecture to perform multiple tasks such as category classification, attribute prediction, and in-shop clothes retrieval. The choice of CNNs, and VGG in particular, was motivated by their strong performance on general image recognition benchmarks and their ability to learn hierarchical feature representations suitable for fine-grained tasks like clothing analysis. The evaluation metrics used include top-k accuracy for classification and retrieval tasks, mean average precision (mAP) for attribute prediction, and Recall@k for retrieval effectiveness. The model demonstrated substantial improvements over prior approaches, showcasing the importance of rich annotations and deep models in fashion-related computer vision tasks.

B. *Fashion Product Classification through Deep Learning and Computer Vision* [2]

This paper presents a practical system developed in collaboration with Adidas AGTM for automatically classifying various garment features. The system aims to recognize logos, stripes, colors, neck and sleeve shapes, and material types from rendered product images. To address the diversity of tasks, a hybrid approach is used, combining standard computer vision techniques (e.g., template matching for small logos and edge-based methods for stripes) with deep learning models for more complex features like logo classification, prints, and material recognition. Specifically, the deep learning modules rely on fine-tuned VGG19 convolutional neural networks, chosen for their balance of simplicity and strong performance compared to alternatives like ResNet or Inception. These CNNs were pre-trained on ImageNet and adapted via transfer learning. Evaluation was conducted using standard metrics such as classifica-

tion accuracy, confusion matrices, and in some cases precision and recall, with the VGG19-based classifiers achieving up to 96% accuracy for sleeve and neck shape detection and 94.4% for prints. This work demonstrates the effectiveness of using a modular deep learning approach combined with traditional vision methods for robust apparel feature classification in real-world industry settings.

C. Deep Learning for Clothing Style Recognition Using YOLOv5 [4]

This paper explored clothing style recognition using the YOLOv5s, a lightweight and efficient one-stage object detection model. The study focused on classifying clothing styles into five categories: plaid, plain, block, horizontal, and vertical. The authors selected YOLOv5s for its small model size and high inference speed, making it well-suited for real-time applications and deployment on resource-constrained platforms. The model was trained and tested using a dataset compiled from DeepFashion, DeepFashion2, and online sources, totaling over 5,000 augmented images. The experimental setup utilized Google Colab with PyTorch and incorporated data augmentation (e.g., Mosaic) to enhance generalization. Evaluation metrics included average precision (AP), mean average precision (mAP), precision, recall, F1-score, model size, and frames per second (FPS). The results demonstrated that YOLOv5s outperformed other models like Faster R-CNN and YOLOv3-tiny, achieving a mAP of 99.1%, precision and recall of 0.97–0.98, and real-time detection speed (40 FPS). This confirmed the model's effectiveness and efficiency for clothing style recognition tasks.

III. BACKGROUND

A. Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are a class of deep learning models that have become the standard for image classification and computer vision tasks. They are designed to automatically and adaptively learn spatial hierarchies of features through backpropagation by using multiple building blocks, such as convolution layers, pooling layers, and fully connected layers.

The core idea behind CNNs is the convolution operation, which applies a set of learnable filters to input images to extract local features. These filters slide over the input data and capture patterns such as edges, textures, or more complex structures as the depth increases. Pooling layers are used to downsample the spatial dimensions, reducing the computational load and helping to control overfitting. Finally, fully connected layers interpret the extracted features and produce class probabilities.

CNNs are particularly effective in vision tasks due to their ability to exploit the 2D structure of images and their invariance to translation and distortion.

B. VGG Networks

VGG is a deep CNN architecture proposed by Simonyan and Zisserman in 2014 [5]. It is known for its simplicity and

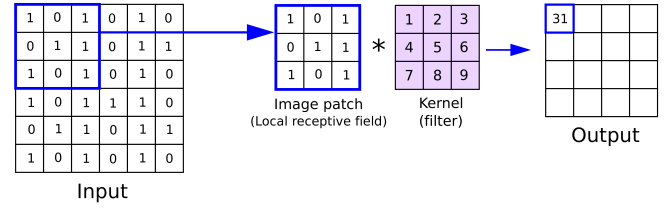


Fig. 1. CNN filter.

uniform architecture. The key characteristic of VGG is its use of small 3×3 convolution filters and a consistent structure where the depth of the network is increased by adding more convolutional layers.

Two of the most popular versions of the VGG architecture are VGG-16 and VGG-19, which contain 16 and 19 weight layers, respectively. Despite its relatively simple design, VGG achieved state-of-the-art performance on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) and has become a widely used baseline model for image classification tasks.

However, one major limitation of VGG networks is their large number of parameters, which makes them computationally expensive and memory intensive.

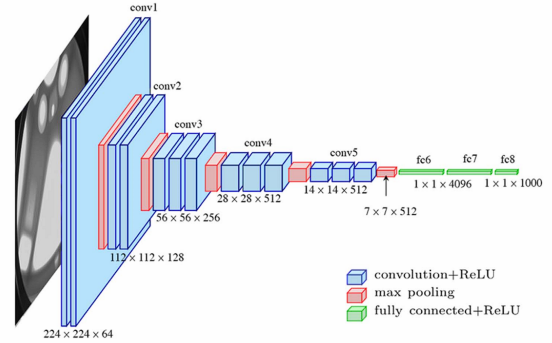


Fig. 2. VGG16 model architecture.

C. ResNet

Residual Networks (ResNet), introduced by He et al. in 2015 [6], address the problem of vanishing gradients in very deep networks. ResNet introduces the concept of residual learning through the use of skip connections (also known as shortcut connections), which allow the gradient to flow directly through the network layers.

The basic building block of ResNet is the residual block, which learns a residual mapping instead of the direct mapping.

ResNet architectures, such as ResNet-50 and ResNet-101, have demonstrated excellent performance on image classification benchmarks and are widely used in practice due to their robustness and scalability.

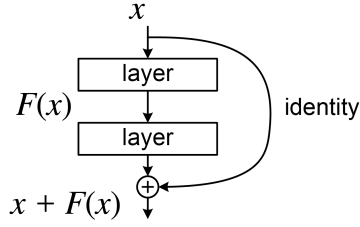


Fig. 3. Example of a Res block.

IV. DATA EXPLORATION

A. Dataset

We worked with 3 datasets for this project:

- Fashion MNIST dataset: a light dataset where the image are 28x28 and in grayscale, we worked on it briefly but it's complexity is too low and not representative on what we think users input on a marketplace like Vinted or SlowFashion. The dataset is already divided in training and testing data. in the training data there is 60K images divided equally among 10 categories.
- Clothing Dataset (Full, High Resolution), kaggle: this dataset include 4846 items, after deletion of the corrupted images, divided in 20 categories one of which being unknown so we removed it. The division was not equal however with t-shirt representing 19% of the dataset, you can find the whole repartition in tab.I.

TABLE I
NUMBER OF ITEMS PER CATEGORY CLOTHING DATASET

Category	Count	Category	Count	Category	Count
T-Shirt	940	Shirt	350	Polo	115
Longsleeve	664	Dress	336	Undershirt	107
Pants	648	Outwear	290	Blazer	102
Shoes	411	Shorts	287	Hoodie	94
Hat	164	Skirt	144	Other	62
Body	62	Top	42	Blouse	21
Skip	7				

- SlowFashion Dataset: Although referred to as a dataset, the SlowFashion data is essentially a list of web links that require parsing to retrieve the actual information. This structure introduces challenges, as the parsing process is not clearly defined and has led to several errors: of the 2,691 available links, 168 main categories were labeled as 'unknown', 62 as 'other', and 3 were incorrectly parsed. Similar issues were observed in the subcategory labels, though without the 'other' class. Furthermore, the dataset suffers from a pronounced class imbalance, as illustrated in Tables II and III. Since the images are hosted online rather than stored locally, a variable number of links were or inaccessible, contributing to missing image data. The character ('-') is used inconsistently in the dataset, sometimes as a replacement for spaces, sometimes to separate hierarchical sections, and occasionally as part of compound words. As a result, some category names

appear incomplete or truncated (e.g., 'T' instead of 'T-Shirt') we're sorry for that and will talk more about our recommendation in the conclusion. The dataset might look large but there is often multiple pictures of the same item with different angles. Pictures also have a white background, often

TABLE II
NUMBER OF ITEMS PER MAIN CATEGORY SLOW FASHION

Category	Count	Category	Count	Category	Count
Clothing	1402	accessories	320	Shoes	308
Apparel	192	Bags	135	Jewellery	101
Other	62	Bad Parsing	3		

TABLE III
NUMBER OF ITEMS PER SUB CATEGORY

Category	Count	Category	Count	Category	Count
Coats	247	Shirts	231	Dresses	212
Jackets	188	Ties	167	High	130
Tops	119	Trousers	119	Skirts	95
Jeans	94	Knitwear	79	Suits	66
Accessories	62	Shoulder	59	Hats	47
Hair	40	Scarves	38	Sneakers	32
Bracelets	32	Sweatshirts	32	Briefcases	27
Chain	24	Belts	23	Casual	20
Trainers	19	Swimwear	19	Sunglasses	19
Boots	18	Oxford	18	Wallets	15
Athletic	15	Loafers	14	Lace	13
Polo	13	Belt	12	Watches	12
Sweaters	12	Sandals	11	Brogues	10
Formal	9	Shorts	9	Jumpsuits	8
Weekend	8	Turtle	8	Flats	6
Derby	6	Ankle	6	Eyewear	6
Lingerie	6	Card	5	Chukka	5
Tie	5	Boat	5	Rings	5
Handbags	5	T	4	Sweats	4
Cufflinks	4	Necklaces	3	Bad Parsing	3

V. APPROACH DESCRIPTION

Our approach is structured around the progressive development and evaluation of increasingly sophisticated image classification models tailored to fashion category recognition. The primary goal was to handle class imbalance, improve generalization, and adapt existing models to work with real-world second-hand clothing images.

We began by prototyping simple convolutional neural networks (CNNs) inspired by earlier coursework to establish baselines. These were intentionally lightweight to facilitate rapid iteration and debugging. We evaluated their performance and limitations, especially in the presence of significant class imbalance.

We then transitioned to deeper models, focusing on the VGG family of architectures (VGG11, VGG16, and VGG19). Observing limited improvements with larger models, we created a custom VGG9 by reducing layers from VGG11 to maintain depth without overfitting. Despite these efforts, models trained from scratch underperformed.

To address this, we adopted transfer learning by leveraging PyTorch's pretrained VGG models trained on ImageNet. We

froze all convolutional layers except for the final Conv5 block and allowed the fully connected layers to fine-tune to our data. This approach preserved valuable feature extraction capabilities while adapting the classifier to our specific task. Class imbalance was mitigated using a weighted cross-entropy loss function, where weights were computed inversely to class frequency.

We also explored traditional machine learning models (e.g., decision trees) for benchmarking purposes. Interestingly, some shallow models outperformed deep networks that had not been pretrained or properly regularized, emphasizing the importance of initialization and training strategies.

Due to time and complexity constraints, we did not pursue object detection using YOLO, though it was initially considered, particularly inspired by prior work using YOLOv5 for fashion style classification [4].

VI. EXPERIMENT DESIGN

Our experimental setup was designed to ensure robust and reproducible evaluation of the various models and configurations across datasets of varying difficulty.

We employed 3-fold cross-validation and 0 epochs for most experiments to assess generalization performance and reduce the risk of overfitting, particularly in the presence of limited or imbalanced data. For Fashion-MNIST, 10-fold cross-validation was used due to its smaller size and balanced structure, enabling more fine-grained performance assessment.

Training was conducted using the Adam optimizer with learning rates adjusted based on early observations: initial experiments used a learning rate of 10^{-3} , later reduced to 10^{-4} to improve convergence stability.

To evaluate the impact of data diversity, we applied standard data augmentation techniques such as random horizontal flips, rotation, resizing, and color jitter. These were tested across CNN-based models to determine their effect on generalization, but in several cases led to worse performance.

For deep architectures, we compared both randomly initialized and pretrained models, the latter significantly outperforming the former. For pretrained VGG models, the final classifier layer was replaced with a fully connected layer corresponding to the number of classes, and only the upper convolutional layers (conv5) and classifier (fc6, fc7, fc8, fig.2) were fine-tuned.

A. Metrics

We used a combination of metrics to evaluate model performance:

- Accuracy: the primary metric used, providing a straightforward measure of overall classification correctness.
- Macro-averaged F1 score: gives equal weight to all classes, useful for evaluating performance on minority classes.
- Weighted F1 score: adjusts for class frequency, reflecting real-world distribution.

- Confusion matrix: offers detailed insights into which classes are frequently misclassified and the nature of those misclassifications.

Based on consultations with the company, a validation accuracy of at least 70% was established as the minimum acceptable performance threshold for deployment or further consideration.

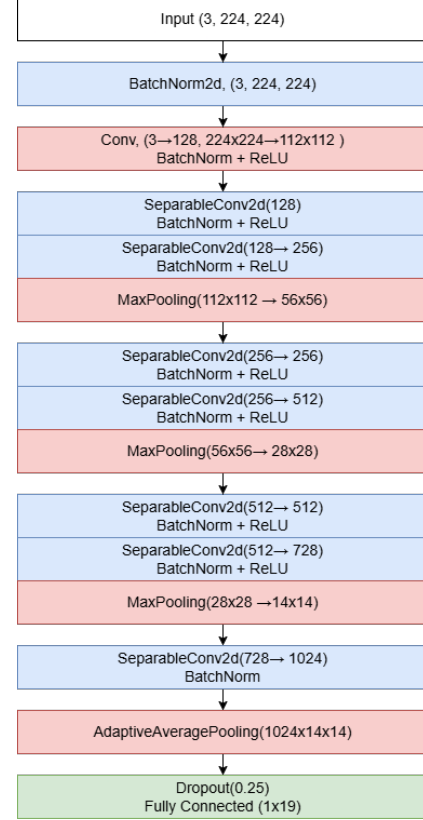


Fig. 4. Model architecture inspired by Assignment 1

VII. RESULT ANALYSIS

Suitable presentation of the obtained results and insightful analysis of the results.

From our experiments on the Fashion-MNIST dataset (tab.IV), it quickly became apparent that this benchmark is relatively easy to perform well on—even a simple decision tree achieved an accuracy of 80%. This suggests that the dataset may not be sufficiently challenging for evaluating more sophisticated models. We also observed that data augmentation did not improve performance and in some cases led to worse results. This may be due to the nature of the dataset; for instance, all shoe images are oriented in the same direction, so applying random flips could introduce distortions it will never see in the testing set and confuse the model. Lastly, a basic CNN architecture with just two convolutional layers and one fully connected layer outperformed all other tested models. This reinforces the established understanding that convolutional neural networks are well-suited for image classification tasks, even in their simplest form.

Model	Accuracy	F1 macro	F1 weighted
Decision Tree	0.80	0.80	0.80
Decision Tree Data Aug.	0.69	0.69	0.69
MLP (64, 32)	0.87	0.87	0.87
Simple CNN	0.92	0.92	0.92
Simple CNN Data aug	0.90	0.90	0.90

TABLE IV
PERFORMANCE ON FASHION MNIST

From the experiments on the Clothing dataset (tab.V), it is evident that pretrained and fine-tuned models significantly outperform those trained from scratch. Even without fine-tuning, pretrained models consistently achieved better performance than their randomly initialized counterparts. Among the architectures tested, ResNet50 yielded poor results. The confusion matrix suggests that its predictions were almost random, indicating underfitting. This may be attributed to insufficient training time, although its per-iteration training time was the longest—approximately 20 seconds—compared to 11 seconds for VGG19 and 7 seconds for our SmallCNN.

Initially, we expected that smaller CNNs might have an advantage, as our custom model outperformed VGG16. However, this result may be due to the longer training time and greater number of parameters required for larger models to converge. Notably, the decision tree delivered surprisingly strong performance given its simplicity, serving as a strong baseline.

We also observed that data augmentation did not improve results. In fact, it led to a drop of about 10 percentage points in accuracy for the SmallCNN model. This suggests that the augmentations applied may have distorted key features in the images or were poorly aligned with the nature of the dataset.

We attempted to enhance the performance of our SmallCNN by adding an additional fully connected layer at the end, taking inspiration from the VGG architecture, which includes three fully connected layers in its design. However, this modification significantly reduced the model’s performance—cutting the accuracy roughly in half. This outcome highlights that increasing model complexity does not always lead to better results.

As shown in the confusion matrix (Fig. 5), class 16 (T-shirt) is frequently misclassified as class 10 (Polo). This confusion is understandable given the visual similarity between the two categories. However, it also highlights an area where the model requires further refinement. Additionally, when examining the predictions for class 10 (Polo), we observe that many of them are actually T-shirts, indicating an imbalance in classification performance. One potential solution to mitigate this issue could be to increase the number of Polo instances in the training set, helping the model to better differentiate between these two closely related classes.

The accuracy-over-epoch plot (Fig. 6) suggests that the VGG19 model could still benefit from further fine-tuning. However, the potential for improvement appears limited, as the training accuracy has begun to plateau, indicating that the model may be approaching its performance ceiling on the

current dataset.

Model	Accuracy	F1 macro	F1 weighted
Decision tree	0.25	0.14	0.25
Small CNN	0.61	0.40	0.59
Small CNN Data augment.	0.50	0.34	0.51
Small CNN more fc	0.30	0.22	0.30
VGG9	0.01	0.00	0.00
VGG11	0.16	0.03	0.08
VGG11 pretrained unfreeze	0.48	0.36	0.51
VGG16	0.01	0.00	0.00
VGG16 pretrained	0.36	0.16	0.33
VGG16 pretrained unfreeze	0.79	0.53	0.68
VGG19	0.09	0.07	0.09
VGG19 pretrained unfreeze	0.80	0.54	0.69
ResNet50	0.07	0.05	0.05

TABLE V
PERFORMANCE ON THE CLOTHING DATASET (FULL, HIGH RESOLUTION)

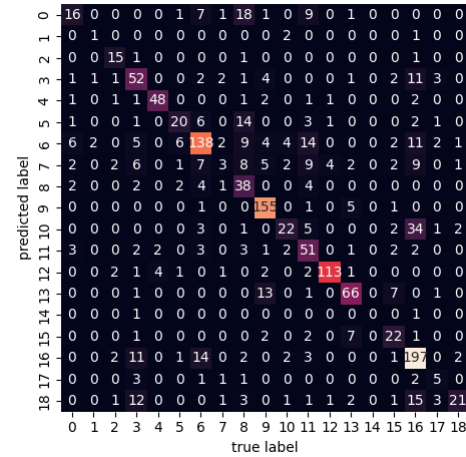


Fig. 5. VGG19 pretrained unfreeze confusion matrix

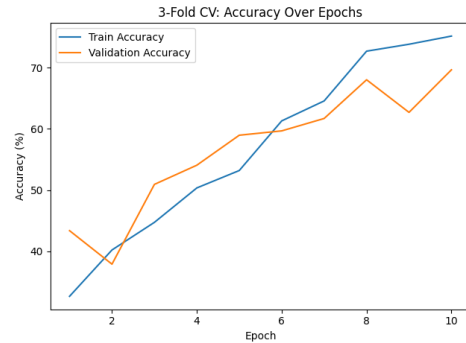


Fig. 6. VGG19 pretrained unfreeze average learning curve

Finally we tried some of our best performing model on the SlowFashion Dataset. The VGG19 pretrained on ImageNet and Clothing Dataset gave the best results at 0.70 (tab.VI) which is also just what we discussed with the company as objective. The Small CNN also gave bad results even considering the complexity of the task. When we saw some of the images we even we didn’t know how to classify them due to there unfamiliar appearance and the camera angles.

Model	Accuracy
Small CNN fine tuned	0.18
VGG19 fine tuned	0.70

TABLE VI
PERFORMANCE ON THE SLOWFASHION DATASET

VIII. CONCLUSION

Concise summary of what has been done, the obtained results and clear recommendations.

In this project, we explored various deep learning architectures for clothing classification. We experimented with custom convolutional neural networks (CNNs), applied data augmentation techniques, and leveraged transfer learning using pretrained models. After identifying our best-performing models, we fine-tuned them on the SlowFashion dataset.

Among all tested approaches, the pretrained VGG19 model—initially trained on ImageNet and further fine-tuned on a clothing dataset—achieved the highest accuracy on the SlowFashion dataset, reaching our target performance of 70%. Our custom small CNN also performed reasonably well, especially considering its simplicity and low computational cost.

Based on our results, we recommend using the VGG19 architecture pretrained on ImageNet and fine-tuned on a dataset with clothing images similar to those in the target application. Notably, the datasets we used lacked categories such as bags, and many images in SlowFashion had uniform white backgrounds. To improve performance, further fine-tuning on more representative datasets like the Fashion Product Images Dataset is advised. We also recommend using a weighted loss function combined with a lower learning rate to mitigate the effects of class imbalance and improve model stability during training.

We also highlight the importance of dataset quality. Increasing the dataset size would likely improve model generalization. Additionally, we recommend providing cleaner, standardized annotations—ideally in a structured CSV file—and avoiding ambiguous or inconsistent formatting (e.g., using the same character for space, hyphen, and category separation).

Lastly, we did not evaluate top-k accuracy in this work, but doing so could offer practical benefits. For instance, if top-3 accuracy is very high, the user interface could be designed to suggest the three most likely categories, improving usability even when the top-1 prediction is incorrect.

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